

EXPERIMENT - 4

Protocols Learnings in detail using Wireshark filter and flow Graph

AIM :

To capture network traffic using Wireshark, list the various network protocols in operation, and analyze at least five protocols (TCP, UDP, ARP, ICMP, and HTTP) in detail using display filters and flow graph.

PROCEDURE :

1. Capturing Packets Using Wireshark

1. Open Wireshark.
2. Select the active network interface (Ethernet / Wi-Fi).
3. Click Start Capture.
4. Perform network activities such as:
 - Opening a website
 - Pinging a host
 - Downloading a file
5. Stop the capture after sufficient packets are collected.

2. Applying Display Filters

Use protocol-specific filters in the Wireshark display filter bar to isolate packets.

Protocol	Display Filter
TCP	<code>tcp</code>
UDP	<code>udp</code>
ARP	<code>arp</code>

ICMP `icmp`

HTTP `http`

3. Using Flow Graph

1. Select any captured packet.
2. Go to Statistics → Flow Graph.
3. Choose TCP Flow / UDP Flow.
4. Click OK to visualize packet exchange between source and destination.

Explanation of Protocols

1. TCP (Transmission Control Protocol)

- TCP is a connection-oriented and reliable transport layer protocol.
- It uses three-way handshake (SYN, SYN-ACK, ACK) to establish a connection.
- Ensures error-free delivery, flow control, and congestion control.

Wireshark Filter: `tcp`

Observation in Flow Graph:

- Displays SYN → SYN-ACK → ACK
- Shows data transfer and connection termination (FIN/ACK)

2. UDP (User Datagram Protocol)

- UDP is a connectionless protocol.
- It provides fast transmission without reliability or error checking.
- Commonly used for DNS, VoIP, video streaming.

Wireshark Filter: `udp`

Observation in Flow Graph:

- No handshake
- One-way or bidirectional packet flow without acknowledgments

3. ARP (Address Resolution Protocol)

- ARP maps IP addresses to MAC addresses in a local network.
- Uses broadcast request and unicast reply mechanism.

Wireshark Filter: `arp`

Observation in Flow Graph:

- ARP Request: "Who has IP?"
- ARP Reply: "IP is at MAC address"

4. ICMP (Internet Control Message Protocol)

- ICMP is used for error reporting and diagnostics.
- Commonly used by the ping command.

Wireshark Filter: `icmp`

Observation in Flow Graph:

- Echo Request
- Echo Reply
- Used to test reachability

5. HTTP (Hyper Text Transfer Protocol)

- HTTP is an application layer protocol used for web communication.
- Works over TCP, typically on port 80.
- Uses request-response model.

Wireshark Filter: `http`

Observation in Flow Graph:

- HTTP GET request from client
- HTTP response from server (200 OK)

OUTPUT :

ARP Packet Analysis (Frame 122)

No.	Time	Source	Destination	Protocol	Length	Info
3237	54.521112	Dell_35:f0:54	Broadcast	ARP	60	Who has 172.16.10.103? Tell 172.16.9.190
3238	54.524933	Sophos_cf:be:45	Broadcast	ARP	60	Who has 172.16.11.63? Tell 172.16.8.1
3239	54.535431	Dell_34:d4:9e	Broadcast	ARP	60	Who has 172.16.8.86? Tell 172.16.8.118
3241	54.583912	EliteGroupCo_14:72:...	Broadcast	ARP	60	Who has 172.16.10.174? Tell 172.16.10.172
3246	54.617410	EliteGroupCo_14:8a:...	Broadcast	ARP	60	Who has 172.16.10.174? Tell 172.16.10.139
3248	54.646927	EliteGroupCo_14:72:...	Broadcast	ARP	60	Who has 172.16.10.210? Tell 172.16.10.172
3260	54.844935	Sophos_cf:be:45	Broadcast	ARP	60	Who has 172.16.11.162? Tell 172.16.8.1
3261	54.858105	Intel_ac:2a:42	Broadcast	ARP	60	Who has 169.254.169.254? Tell 172.16.11.36
3262	54.859768	EliteGroupCo_12:27:...	Broadcast	ARP	60	Who has 172.16.10.174? Tell 172.16.10.203
3276	55.252975	Dell_35:f0:54	Broadcast	ARP	60	Who has 172.16.10.103? Tell 172.16.9.190
3298	55.488640	0a:e0:af:ca:01:9e	Broadcast	ARP	60	Who has 172.16.8.1? Tell 172.16.8.25
3299	55.500501	EliteGroupCo_15:ee:...	Broadcast	ARP	60	Who has 172.16.10.174? Tell 172.16.10.175
3302	55.518911	EliteGroupCo_15:ee:...	Broadcast	ARP	60	Who has 172.16.10.174? Tell 172.16.10.178
3307	55.586965	EliteGroupCo_14:72:...	Broadcast	ARP	60	Who has 172.16.10.174? Tell 172.16.10.172
3308	55.586965	EliteGroupCo_14:72:...	Broadcast	ARP	60	Who has 172.16.10.210? Tell 172.16.10.172
3309	55.615248	EliteGroupCo_14:8a:...	Broadcast	ARP	60	Who has 172.16.10.174? Tell 172.16.10.139
3332	55.958293	MicroStarInt_ad:3f:...	Broadcast	ARP	60	Who has 172.16.10.103? Tell 172.16.8.8
3335	56.009805	MicroStarInt_ad:3e:...	Broadcast	ARP	60	Who has 172.16.10.66? Tell 172.16.8.23

Frame 122: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF...
Ethernet II, Src: EliteGroupCo_15:ee:12 (88:ae:dd:15:ee:12), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)

Address Resolution Protocol: Protocol

Packets: 3380 · Displayed: 834 (24.7%)

Profile: Default

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TCP Packet Analysis (Frame 595)

No.	Time	Source	Destination	Protocol	Length	Info
581	8.820161	184.28.108.196	172.16.10.38	TLSv1.3	390	Application Data
582	8.820890	172.16.10.38	184.28.108.196	TCP	54	11540 → 443 [FIN, ACK] Seq=419 Ack=4422 Win=261888 Len=0
583	8.823900	184.28.108.196	172.16.10.38	TLSv1.3	390	Application Data
584	8.823346	172.16.10.38	184.28.108.196	TCP	54	11541 → 443 [FIN, ACK] Seq=722 Ack=879 Win=261888 Len=0
585	8.836674	184.28.108.196	172.16.10.38	TLSv1.3	78	Application Data
586	8.836674	184.28.108.196	172.16.10.38	TCP	60	443 → 11541 [FIN, ACK] Seq=903 Ack=723 Win=63744 Len=0
587	8.836726	172.16.10.38	184.28.108.196	TCP	54	11541 → 443 [RST, ACK] Seq=723 Ack=903 Win=0 Len=0
588	8.851832	184.28.108.196	172.16.10.38	TLSv1.3	78	Application Data
589	8.851832	184.28.108.196	172.16.10.38	TCP	60	443 → 11540 [FIN, ACK] Seq=4446 Ack=420 Win=64128 Len=0
590	8.851869	172.16.10.38	184.28.108.196	TCP	54	11540 → 443 [RST, ACK] Seq=420 Ack=4446 Win=0 Len=0
591	8.856354	184.28.108.196	172.16.10.38	TLSv1.3	390	Application Data
592	8.856690	172.16.10.38	184.28.108.196	TCP	54	11542 → 443 [FIN, ACK] Seq=419 Ack=4422 Win=261888 Len=0
593	8.870064	184.28.108.196	172.16.10.38	TLSv1.3	78	Application Data
594	8.870064	184.28.108.196	172.16.10.38	TCP	60	443 → 11542 [FIN, ACK] Seq=4446 Ack=420 Win=64128 Len=0
595	8.870103	172.16.10.38	184.28.108.196	TCP	54	11542 → 443 [RST, ACK] Seq=420 Ack=4446 Win=0 Len=0
600	8.912010	184.28.108.196	172.16.10.38	TCP	60	[TCP Retransmission] 443 → 11541 [FIN, ACK] Seq=903 Ack=723 Win=63744 Len=0
619	9.146099	184.28.108.196	172.16.10.38	TCP	78	[TCP Spurious Retransmission] 443 → 11541 [FIN, PSH, ACK] Seq=879 Ack=723 Win=63744 Len=24
638	9.628724	184.28.108.196	172.16.10.38	TCP	78	[TCP Spurious Retransmission] 443 → 11541 [FIN, PSH, ACK] Seq=879 Ack=723 Win=63744 Len=24

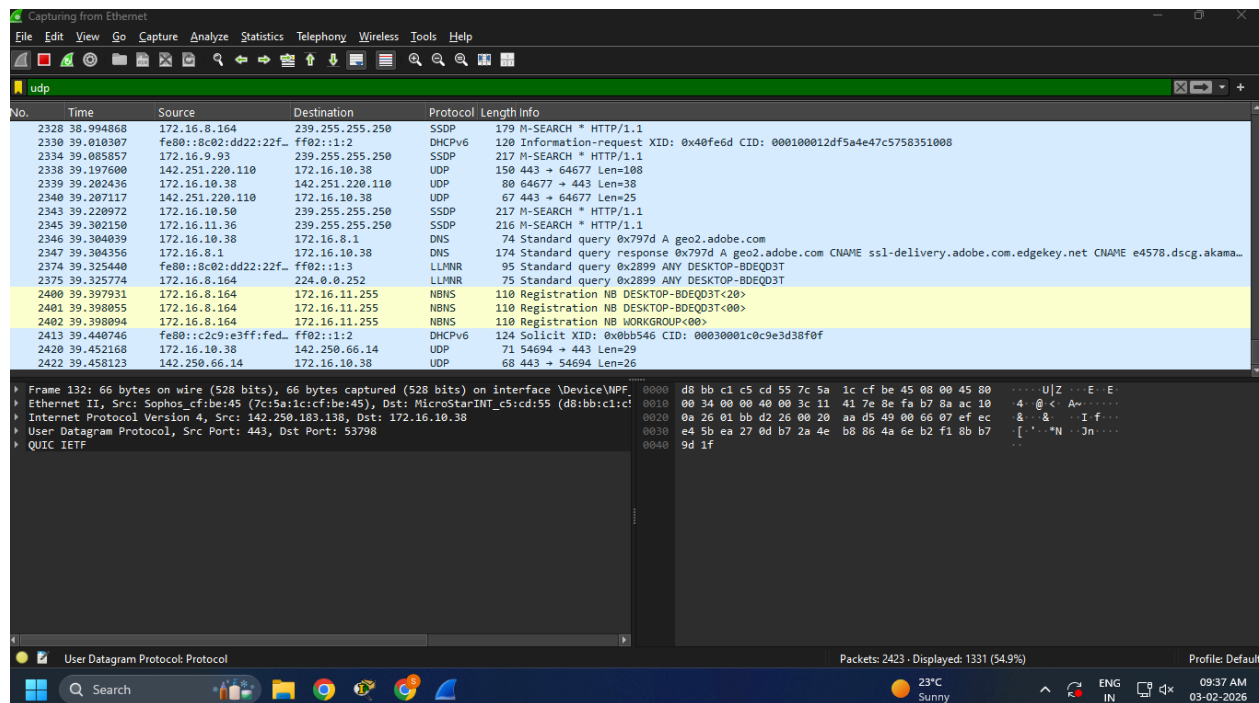
Frame 133: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF...
Ethernet II, Src: MicroStarInt_c5:cd:55 (d8:bb:c1:c5:cd:55), Dst: Sophos_cf:be:45 (7c:5a:1c:c1:c5:cd:55)
Internet Protocol Version 4, Src: 172.16.10.38, Dst: 142.251.43.138
Transmission Control Protocol, Src Port: 44793, Dst Port: 443, Seq: 0, Len: 0

Transmission Control Protocol: Protocol

Packets: 649 · Displayed: 109 (16.8%)

Profile: Default

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RESULT :

Network traffic was successfully captured and analyzed using Wireshark. Various protocols such as TCP, UDP, ARP, ICMP, and HTTP were identified and filtered using display filters. The working of each protocol was studied in detail using packet analysis and flow graphs. The experiment demonstrated how different protocols operate at various layers of the network and how Wireshark can be used as an effective protocol analysis tool